Plan — Phase-Shaped Moving-Precursor Dynamic Capture

READY → moving handoff → frozen downstream → strict K6 (structured CEM, no TD3)

HyMeKo coin-delivery campaign (`recovery/coin-r9-causal-residual-delivery`)

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Abstract

Stage 1A solved `HOME_V1`→`READY` analytically. Stage 1B closes the last, contact-rich transition into the frozen downstream. The decisive reframing: the cradle handoff $(q^\star, \dot{q}^\star, \tau^\star)$ is *not* a global capture target but a **phase-point of a moving orbit** — q is the position and $\dot{q}$ the *tangent* of the same trajectory, $\tau^\star$ the control-history memory. Hitting all three jointly under a terminal-state objective is ill-conditioned (position *xor* velocity, measured). Instead we arrive *moving along the orbit's tangent*: a quintic terminal segment drives the tips from `READY` to a backward-tangent precursor $q_{\text{pre}}(\Delta t) = q^\star - \Delta t\, \dot{q}^\star$ with terminal velocity $\approx \dot{q}^\star$; the τ preload ramps *during* that moving segment (applying $\tau^\star$ afterward accelerates the arm out of the deliverable velocity band); and a short residual under a low-dimensional structured CEM closes the margin. The frozen APPROACH→HANDOFF_RESET→R2→coast then generates the coupling itself. Positive control already in hand: **strict K6 reproduced 3/3 independent planner seeds** (min d_{tz} 2.0/9.7/10.9 mm, safe, distinct parameters), no state edit, no snapshot injection. This plan formalises it and freezes the traces. *Join the orbit; do not reconstruct the terminal.*

1 Scope and goal

Goal. A deterministic structured planner that, from the frozen `READY` shell, drives a phase-shaped moving precursor into the frozen downstream and delivers strict K6 — reproducibly across ≥ 3 planner seeds, teacher-free, no state edit. **Chain:**

HOME_V1 → frozen analytic transit → READY → phase-shaped moving precursor (quintic + during-segment preload + short residual) → frozen APPROACH → HANDOFF_RESET → R2 → strict K6.

Non-goals. No TD3 (this is the positive control that gates it). No modification of the frozen downstream, the analytic transit, or the physics/monitor. No re-characterisation of a different deliverable basin unless this fork regresses (that is fork B, deferred).

2 The frozen contract (must hold at every step)

- Physics + motion contract + K6 monitor unchanged; every torque through the existing `step_ablation`→governor stack; no raw actuator writes, no teleport, no coin-state edit, no hidden force in the deployed chain. State-editing appears *only* as a characterisation tool (basin/tube mapping), never deployed.
- Frozen downstream imported as-is: `HandoffResetTemporalController` (APPROACH → HANDOFF_RESET → R2 α=0.15 → coast), `velocity_rollout`, `delivery_success`. `git diff` empty on them.

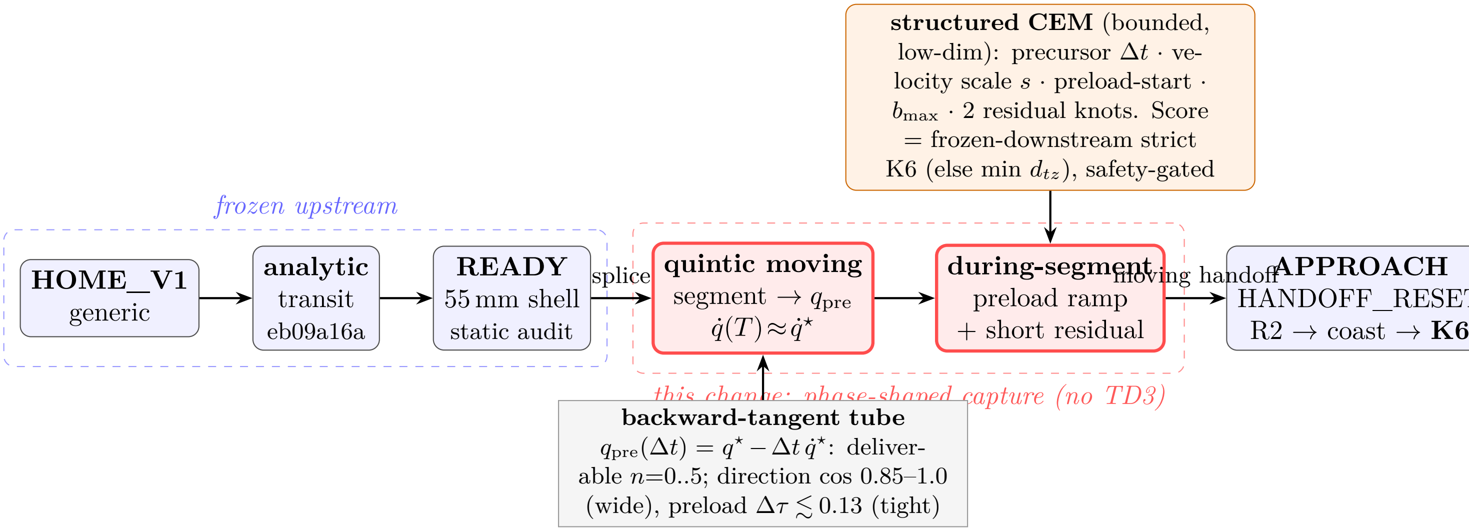

Figure 1: The phase-shaped capture (red) splices the frozen analytic transit (blue) into the frozen downstream (blue) by a quintic moving segment onto the backward-tangent precursor tube; the structured CEM (orange) searches only bounded structural parameters. The cradle handoff is a phase-point of the moving orbit, not a static target.

- Frozen upstream: the Stage 1A analytic transit (`planar_geometric_approach`, committed `eb09a16a`) produces `READY`; unchanged. `HOME_STATE_V1_GENERIC` retained.
- `s4/s7` validation-only; `f1--f4 SEALED`. No tag moved.

3 Affected files

CORE.YAML items touched: none. New code under `hymeko_rl/` + `tests//reports//docs/plans/`.

- `hymeko_rl/coin_delivery/theta_option/moving_precapture.py` — `HandoffReference` (the cradle phase-point + backward-tangent precursor), `QuinticSegment` (position+velocity BVP), `CaptureParams` + `CaptureSearchSpec`, `PhaseShapeCapture` (quintic track + during-segment preload ramp + residual, through the governed servo), `FrozenDownstream` (thin adapter over the frozen controller), `plan_capture` (the structured CEM), `execute_home_to_k6` (full chain).
- `hymeko_rl/experiments/coin_kinetic_moving_precapture.py` — 3-seed reachability gate + boundary tests + saved traces + provenance JSON.
- `hymeko_rl/tests/test_moving_precapture.py` — unit + integration + boundary + perf.
- `reports/2026-07-28-moving-precapture-dynamic-handoff.md` — the report.

Read-only reuse: `planar_geometric_approach` (READY), `teacher_bank` (cradle), `kinetic_*` frozen downstream, `env/motion_contract` (governor).

4 Interface changes

All additive.

- `HandoffReference.from_cradle(cradle)` $\to$ `HandoffReference` with $(q^\star, \dot q^\star, \tau^\star)$ and `precursor(n, dt)` $\to$ `q_pre`. **Post:** deterministic; `precursor(0,_)`$= q^\star$.
- `QuinticSegment(q0,v0,qf,vf,T)` with `eval(t)` $\to$ `(q,v)`. **Post:** boundary conditions met $(q(0), \dot q(0), q(T), \dot q(T);$ $\ddot q(0)=\ddot q(T)=0)$, verified by test.

- `PhaseShapeCapture(ready, ref, params).roll()` $\to$ `CaptureOutcome` — governed-servo roll; reports the terminal snapshot, direction cosine to $\dot{q}^\star$, velocity scale, $\Delta\tau$ to $\tau^\star$, contact, safety. **Post:** deterministic given `(ready, params)`; no state edit.
- `plan_capture(ready, ref, spec, seed, downstream)` $\to$ `(CaptureParams, CaptureResult)` — CEM over the bounded structural vector ($\Delta t, s,$ preload_start$, b_{\max}, 2$ residual knots), scored by the frozen downstream (0 if K6 else min d_{tz}; safety hard-penalised). **Post:** deterministic given seed.

5 Test strategy

Unit: quintic BVP boundary conditions; backward-tangent precursor ($q_{\text{pre}}(0)=q^\star$, monotone recession); `HandoffReference` determinism; `PhaseShapeCapture.roll` determinism + no-state-edit (coin canonical at start); the tube-membership predicate (direction cosine wide / preload tight, from the measured tolerances) as a regression fixture. **Integration:** the full chain from `HOME_V1` delivers strict K6 for at least one seed (`MOVING_PRECAPTURE_TO_HANDOFF`); a naive *post*-segment preload ramp *fails* the velocity-tube (regression proving the during-segment ramp is load-bearing). **Boundary:** `READY_TO_CAPTURE_ONE_STEP_IDENTITY` (offline READY = one online capture step from READY); `TRANSIT_TO_CAPTURE_CONTINUATION`; exactly-one TRANSIT→CAPTURE and one REACH_TO_APPROACH; an off-by-one boundary rejected. **Perf:** a single capture+downstream evaluation wall-bounded; median over 5. **Coverage:** every new public/private function exercised by ≥ 1 new test.

6 Performance budget

Anchors (Apple-Silicon, py3.11.15, mujoco 3.10.0): one capture ($\sim$20 steps) + one downstream rollout ($\sim$80 steps) $\approx$ 0.1 s; CEM $\sim$14 iters $\times$ 44 pop $\approx$ 620 evals $\approx$ 60 s/seed; 3 seeds $\approx$ 3 min, *checkpointed per seed.* **Peak RSS** $<$ **1 GB** (hard cap 16 GB). The reachability run reports peak RSS + wall on exit; unit suite $<$ 120 s.

7 Risk anticipation

Performance-contract preservation. The capture adds no new torque authority: every command passes the same `step_ablation`→`govern_torque` clip (slew, torque, joint/coin-speed caps). The residual is bounded (±slew) and enters inside that clip. **Worst-case / the load-bearing coupling.** $\tau^\star$ ($\|\cdot\|$=2.67) is the *accelerating* torque ($\dot{q}$ jumps 0.66 $\to$ 1.07 in one downstream step). Applying it *after* the moving segment drives velocity to $\sim 2\times$ the deliverable band (measured). Mitigation = ramp the preload *during* the decelerating quintic, with the terminal-velocity scale s compensating; the CEM searches (preload_start, $b_{\max}$, s) jointly. The deliverable tube is tight on preload ($\Delta\tau \lesssim 0.13$) and wide on direction (cos 0.85–1.0) — the plan targets the wide axis loosely and the tight axis via the during-segment ramp. The precursor must keep the coin canonical until legal contact (checked) and stay safe (`peak_qdot`$\leq$ 3, `peak_coin_speed`$\leq$ 1.5). **Reproducibility.** ≥ 3 *independent* seeds must each deliver, and preferably not the same parameter vector (the 3 measured differ: $s \in \{0.08, 0.28, 0.31\}$). Seeds, chosen parameters, and full causal traces are saved.

8 Rollback path

New files only; rollback = delete. The frozen transit + downstream are untouched, so a regression reverts to the Stage 1A commit `eb09a16a` with no side effects.

9 Gate ladder

1. `H_DYN_RECLASSIFIED_AS_LOCAL_CONTINUATION_BASIN` (done, diagnostic): the handoff is a moving phase-point; direction-wide / preload-tight tube; backward-tangent tube deliverable for n=0..5.
2. `MOVING_PRECAPTURE_TO_HANDOFF_PASS`: the phase-shaped chain from `HOME_V1` delivers strict K6 (one seed), safe, no state edit.
3. `HOME_V1_TO_DYNAMIC_HANDOFF_REACHABILITY_PASS`: ≥ 3 independent planner seeds each deliver strict K6, safe, deterministic replay, exactly-one boundary transitions, full provenance + saved traces.
4. *(next, not here)* freeze $\rightarrow$ TD3 (`HOME_V1_START_STRICT_K6`); or fork B if this regresses.

Stop logic. ≥ 3 K6 $\Rightarrow$ save traces, freeze, commit, **HALT** before TD3. Precursor reached but no K6 $\Rightarrow$ fork B (reachable-deliverable-basin, direct downstream score). Do not modify the frozen downstream.

10 Empty-plan-dir hygiene

If abandoned before the module lands, delete `docs/plans/2026-07-28-moving-precapture-dynamic-handoff/` this session.